

CFIP Complete Data Model Reference

All Tables and Relations - REVISED ARCHITECTURE

Version: 2.1 - Financial Domain Redesign

Last Updated: February 2026

Total Tables: 25

Total Relationships: 55+

DOMAIN 1: Identity & User Management

Table: users

Core authentication table. Every user in the system has an entry here.

Column	Type	Description
user_id	UUID	Primary key
username	TEXT	Unique login name (UNIQUE, NOT NULL)
email	TEXT	Unique email address (UNIQUE, NOT NULL)
password_hash	TEXT	Hashed password (bcrypt/argon2) (NOT NULL)
user_role	TEXT	VISITOR OWNER GOV_USER CARRIER_STAFF (NOT NULL)
is_active	BOOLEAN	Account active status (DEFAULT TRUE)
last_login_at	TIMESTAMP	Last successful login
created_at	TIMESTAMP	Account creation (DEFAULT NOW())
updated_at	TIMESTAMP	Last update (DEFAULT NOW())

Relationships: 1:1 → visitors | owners | gov_users | carrier_staff | 1:N → financial_transactions | user_activity_log

Table: visitors

Foreign nationals who request TNAs. Extended profile beyond base user authentication.

Column	Type	Description
visitor_id	UUID	Primary key
user_id	UUID	FK → users.user_id (UNIQUE, NOT NULL)
full_name	TEXT	Legal full name (NOT NULL)
nationality	TEXT	ISO 3166-1 alpha-2 country code (NOT NULL)
date_of_birth	DATE	Date of birth (NOT NULL)
document_type	TEXT	VISA IQAMA PASSPORT (NOT NULL)
document_number	TEXT	Official document ID number (UNIQUE, NOT NULL)
document_expiry	DATE	Document expiration date
mobile	TEXT	Mobile phone number (E.164 format)
emergency_contact	TEXT	Emergency contact phone

bank_account_info	JSON	Bank details for refunds {bank_name, account_number, iban}
max_active_tnas	INTEGER	Maximum concurrent active TNAs (DEFAULT 3)
kyc_status	TEXT	PENDING VERIFIED REJECTED (DEFAULT PENDING)
kyc_verified_at	TIMESTAMP	KYC verification timestamp
created_at	TIMESTAMP	Profile creation (DEFAULT NOW())

Relationships: 1:N → visitor_tnas | tna_issuance_requests

Table: owners

Property owners (individuals or businesses) who register national addresses and rent out sub-addresses.

Column	Type	Description
owner_id	UUID	Primary key
user_id	UUID	FK → users.user_id (UNIQUE, NOT NULL)
owner_type	TEXT	INDIVIDUAL BUSINESS (NOT NULL)
full_name	TEXT	Owner full name (NOT NULL)
business_name	TEXT	Company name (required if owner_type=BUSINESS)
commercial_registration	TEXT	CR number (required if BUSINESS)
national_id	TEXT	National ID number (UNIQUE, NOT NULL)
tax_id	TEXT	Tax identification number
mobile	TEXT	Contact phone number (NOT NULL)
nationality	TEXT	Nationality/country of registration
bank_account_info	JSON	Bank details for payouts {bank_name, account_number, iban, swift}
is_verified	BOOLEAN	Owner verification status (DEFAULT FALSE)
verified_at	TIMESTAMP	Verification timestamp
created_at	TIMESTAMP	Profile creation (DEFAULT NOW())

Relationships: 1:N → owner_national_addresses | bindings (as approver) | 1:1 → owner_accounts

Table: government_agencies

Government departments and their hierarchical structure. Supports RBAC for different agency permissions.

Column	Type	Description
agency_id	UUID	Primary key
agency_code	TEXT	MOFA MOI_PASSPORT MOI_IMMIGRATION MOMRA CUSTOMS (UNIQUE, NOT NULL)
agency_name_en	TEXT	English name of the agency (NOT NULL)
agency_name_ar	TEXT	Arabic name of the agency
parent_agency_id	UUID	FK → government_agencies.agency_id (hierarchical structure)
permissions	JSON	Agency permissions ["view_tna_requests", "approve_tna", "verify_addresses"]
is_active	BOOLEAN	Agency operational status (DEFAULT TRUE)
created_at	TIMESTAMP	Agency registration (DEFAULT NOW())

Relationships: 1:N → gov_users | Self-referencing for hierarchy

Table: gov_users

Government employees with department affiliation and role-based permissions.

Column	Type	Description
gov_user_id	UUID	Primary key
user_id	UUID	FK → users.user_id (UNIQUE, NOT NULL)
agency_id	UUID	FK → government_agencies.agency_id (NOT NULL)
full_name	TEXT	Employee full name (NOT NULL)
employee_id	TEXT	Government employee ID (UNIQUE, NOT NULL)
department	TEXT	Department within agency
position	TEXT	Job title/position
role	TEXT	REVIEWER APPROVER

		ADMIN AUDITOR (NOT NULL)
additional_permissions	JSON	User-specific permission overrides
is_active	BOOLEAN	Employment status (DEFAULT TRUE)
created_at	TIMESTAMP	Profile creation (DEFAULT NOW())

Relationships: 1:N → tna_issuance_requests (as reviewer) | na_sub_addresses (as verifier)

Table: carriers

Logistics companies as business entities. Represents the corporate entity with licensing and API access.

Column	Type	Description
carrier_id	UUID	Primary key
company_name	TEXT	Legal company name (UNIQUE, NOT NULL)
commercial_registration	TEXT	Commercial registration number (UNIQUE, NOT NULL)
license_number	TEXT	Logistics/transportation license number (UNIQUE, NOT NULL)
tax_id	TEXT	Tax identification number
headquarters_address	TEXT	Company headquarters address
contact_email	TEXT	Primary contact email (UNIQUE, NOT NULL)
contact_phone	TEXT	Primary contact phone (NOT NULL)
api_key	TEXT	API authentication key (UNIQUE, NOT NULL)
api_secret_hash	TEXT	Hashed API secret
webhook_url	TEXT	Webhook endpoint for delivery updates
is_verified	BOOLEAN	Carrier verification status (DEFAULT FALSE)
verified_at	TIMESTAMP	Verification timestamp
is_active	BOOLEAN	Operational status (DEFAULT TRUE)
created_at	TIMESTAMP	Registration (DEFAULT NOW())

Relationships: 1:N → carrier_staff | shipments

Table: carrier_staff

Individual employees of carrier companies who interact with the system.

Column	Type	Description
staff_id	UUID	Primary key
user_id	UUID	FK → users.user_id (UNIQUE, NOT NULL)
carrier_id	UUID	FK → carriers.carrier_id (NOT NULL)
full_name	TEXT	Staff member full name (NOT NULL)
employee_id	TEXT	Employee ID within carrier company
position	TEXT	Job title (e.g., DRIVER, DISPATCHER, MANAGER)
mobile	TEXT	Contact phone number
is_active	BOOLEAN	Employment status (DEFAULT TRUE)
created_at	TIMESTAMP	Profile creation (DEFAULT NOW())

Relationships: 1:N → shipments (as assigned handler)

DOMAIN 2: TNA Lifecycle Management

Table: tna_issuance_requests

Workflow tracking for TNA issuance. Supports both moderated (government review) and autonomous modes.

Column	Type	Description
request_id	UUID	Primary key
visitor_id	UUID	FK → visitors.visitor_id (NOT NULL)
request_status	TEXT	PENDING UNDER_REVIEW APPROVED REJECTED ISSUED (NOT NULL)
mode_at_submission	TEXT	MODERATED AUTONOMOUS (NOT NULL)
reviewed_by_gov_user_id	UUID	FK → gov_users.gov_user_id (required if MODERATED)
rejection_reason	TEXT	Reason for rejection if status=REJECTED
supporting_documents	JSON	Array of document URLs [{doc_type, url, uploaded_at}]
created_at	TIMESTAMP	Request submission (DEFAULT NOW())
reviewed_at	TIMESTAMP	Government review completion
issued_at	TIMESTAMP	TNA issuance timestamp

Relationships: 1:1 → visitor_tnas (when approved/issued)

Table: visitor_tnas

The TNA registry. Each TNA is a unique temporary national address code assigned to a visitor.

Column	Type	Description
tna_id	UUID	Primary key
visitor_id	UUID	FK → visitors.visitor_id (NOT NULL)
issuance_request_id	UUID	FK → tna_issuance_requests.request_id (UNIQUE)
tna_code	TEXT	Unique TNA identifier (e.g., TNA-2024-ABC123) (UNIQUE, NOT NULL)
status	TEXT	UNLINKED ACTIVE EXPIRED REVOKED (NOT NULL, DEFAULT UNLINKED)
issued_at	TIMESTAMP	TNA issuance timestamp (NOT

		NULL, DEFAULT NOW())
expires_at	TIMESTAMP	TNA expiration timestamp (NOT NULL)
revoked_at	TIMESTAMP	Revocation timestamp if applicable
revocation_reason	TEXT	Reason for revocation

Relationships: 1:N → bindings | shipments

DOMAIN 3: Physical Property Registry

Table: owner_national_addresses

National addresses registered by property owners. Each address must have proof of ownership before sub-addresses can be created.

Column	Type	Description
na_id	UUID	Primary key
owner_id	UUID	FK → owners.owner_id (NOT NULL)
full_address	TEXT	Complete national address (NOT NULL)
building_number	TEXT	Building number component
street_name	TEXT	Street name
district	TEXT	District/neighborhood
city	TEXT	City name (NOT NULL)
postal_code	TEXT	Postal/zip code
additional_number	TEXT	Additional number (short code)
unit_number	TEXT	Unit number if applicable
latitude	DECIMAL	GPS latitude coordinate
longitude	DECIMAL	GPS longitude coordinate
registry_reference	JSON	National address system reference ID (PDF file)
title_deed_reference	TEXT	Title deed/property deed reference number (REQUIRED)
na_certificate_url	TEXT	URL to national address certificate document
ownership_proof_status	TEXT	PENDING VERIFIED REJECTED (DEFAULT PENDING)
verified_at	TIMESTAMP	Ownership verification timestamp
created_at	TIMESTAMP	Address registration (DEFAULT NOW())

Relationships: 1:N → na_sub_addresses

Table: na_sub_addresses

Subdivisions within a national address (units, apartments, suites). Each sub-address must be government-verified before it can be used in a binding.

Column	Type	Description
sub_address_id	UUID	Primary key
na_id	UUID	FK → owner_national_addresses.na_id (NOT NULL)
suffix_code	TEXT	Sub-address suffix (e.g., APT-101, UNIT-A) (NOT NULL)
label	TEXT	Human-readable label (e.g., Apartment 101)
description	TEXT	Additional description of the sub-address
is_available	BOOLEAN	Whether sub-address is available for binding (DEFAULT TRUE)
is_verified	BOOLEAN	Government verification status (DEFAULT FALSE, REQUIRED for binding)
verified_at	TIMESTAMP	Verification timestamp
verified_by_gov_user_id	UUID	FK → gov_users.gov_user_id (government official who verified)
verification_notes	TEXT	Notes from verification process
created_at	TIMESTAMP	Sub-address creation (DEFAULT NOW())

Relationships: 1:N → bindings | Constraint: UNIQUE(na_id, suffix_code)

DOMAIN 4: Rental Contracts

Table: rentalContract

The core rental contract. Links a TNA to a verified sub-address for a specific time period with owner approval and financial contract.

Column	Type	Description
binding_id	UUID	Primary key
tna_id	UUID	FK → visitor_tnas.tna_id (NOT NULL)
sub_address_id	UUID	FK → na_sub_addresses.sub_address_id (NOT NULL)
rent_contract_id	UUID	FK → rent_contracts.contract_id (UNIQUE)
status	TEXT	PENDING ACTIVE COMPLETED CANCELLED TERMINATED (NOT NULL)
start_at	TIMESTAMP	Binding start timestamp (NOT NULL)
end_at	TIMESTAMP	Binding end timestamp (NOT NULL)
approved_by_owner_id	UUID	FK → owners.owner_id (owner who approved)
approved_at	TIMESTAMP	Owner approval timestamp
termination_reason	TEXT	Reason if status=TERMINATED or CANCELLED
created_at	TIMESTAMP	Binding creation (DEFAULT NOW())
updated_at	TIMESTAMP	Last update (DEFAULT NOW())

Relationships: 1:1 → rent_contracts | 1:N → binding_status_history

Table: rental_status_history

Complete audit trail of all binding status changes for compliance and dispute resolution.

Column	Type	Description
history_id	UUID	Primary key
binding_id	UUID	FK → bindings.binding_id (NOT NULL)
old_status	TEXT	Previous status
new_status	TEXT	New status (NOT NULL)
changed_by_user_id	UUID	FK → users.user_id (who made the change)
change_reason	TEXT	Reason for status change
metadata	JSON	Additional context about the change
changed_at	TIMESTAMP	Status change timestamp (DEFAULT NOW())

Relationships: None (audit table)

DOMAIN 5: Logistics & Delivery

Table: shipments

Package delivery tracking. Links carrier, TNA, and delivery status with real-time updates.

Column	Type	Description
shipment_id	UUID	Primary key
carrier_id	UUID	FK → carriers.carrier_id (NOT NULL)
tracking_number	TEXT	Carrier's tracking number (UNIQUE, NOT NULL)
tna_id	UUID	FK → visitor_tnas.tna_id (delivery destination)
assigned_staff_id	UUID	FK → carrier_staff.staff_id (driver/handler)
status	TEXT	CREATED PICKED_UP IN_TRANSIT OUT_FOR_DELIVERY DELIVERED FAILED RETURNED (NOT NULL)
origin_address	TEXT	Shipment origin address
destination_address_full	TEXT	Complete destination address (resolved from TNA+binding)
estimated_delivery	TIMESTAMP	Estimated delivery timestamp
actual_delivery	TIMESTAMP	Actual delivery timestamp
delivery_signature	TEXT	Proof of delivery signature/reference
delivery_photo_url	TEXT	Photo proof of delivery URL
package_details	JSON	Package description {weight, dimensions, contents}
failure_reason	TEXT	Reason if status=FAILED
created_at	TIMESTAMP	Shipment creation (DEFAULT NOW())
updated_at	TIMESTAMP	Last update (DEFAULT NOW())

Relationships: 1:N → shipment_messages | shipment_status_logs

Table: shipment_status_logs

Real-time tracking log of all shipment status changes with GPS coordinates and timestamps.

Column	Type	Description
log_id	UUID	Primary key
shipment_id	UUID	FK → shipments.shipment_id (NOT NULL)
status	TEXT	Status at this checkpoint (NOT

		NULL)
location	TEXT	Location description
latitude	DECIMAL	GPS latitude
longitude	DECIMAL	GPS longitude
notes	TEXT	Additional notes/details
logged_at	TIMESTAMP	Log entry timestamp (DEFAULT NOW())

Relationships: None (log table)

Table: shipment_messages

Communication thread for each shipment between visitor, carrier, and platform.

Column	Type	Description
message_id	UUID	Primary key
shipment_id	UUID	FK → shipments.shipment_id (NOT NULL)
sender_user_id	UUID	FK → users.user_id (NOT NULL)
message_type	TEXT	TEXT NOTIFICATION ALERT (DEFAULT TEXT)
message_text	TEXT	Message content (NOT NULL)
attachments	JSON	Array of attachment URLs [{url, type, filename}]
is_read	BOOLEAN	Read status (DEFAULT FALSE)
created_at	TIMESTAMP	Message timestamp (DEFAULT NOW())

Relationships: None

DOMAIN 6: Financial System (Financials Service Architecture)

The financial domain implements a Financials Service architecture with configuration-driven pricing, formal rent contracts, double-entry ledger accounting, and structured payout rails.

Table: price_catalog

Configuration-driven pricing module. Allows dynamic pricing for TNA issuance and rental periods without code deployment.

Column	Type	Description
catalog_id	UUID	Primary key
item_type	TEXT	TNA_ISSUANCE RENTAL_DAILY RENTAL_MONTHLY RENTAL_YEARLY SERVICE_FEE (NOT NULL)
item_name	TEXT	Human-readable name (e.g., Standard TNA, Premium Rental)
base_price	DECIMAL(10,2)	Base price in SAR (NOT NULL)
currency	TEXT	Currency code (DEFAULT SAR)
pricing_rules	JSON	Dynamic pricing rules {visitor_type_multiplier, volume_discount, seasonal_rates}
platform_fee_percentage	DECIMAL(5,2)	Platform commission percentage (e.g., 15.00 for 15%)
authority_share_percentage	DECIMAL(5,2)	Government authority share percentage (e.g., 5.00 for 5%)
is_active	BOOLEAN	Pricing active status (DEFAULT TRUE)
effective_from	TIMESTAMP	When pricing takes effect (NOT NULL)
effective_until	TIMESTAMP	When pricing expires (NULL = indefinite)
created_at	TIMESTAMP	Catalog entry creation (DEFAULT NOW())
updated_at	TIMESTAMP	Last pricing update (DEFAULT NOW())

Relationships: Referenced by: rent_contracts | financial_transactions

Table: rent_contracts

Formalizes the financial agreement for each binding. Defines revenue split structure and rental terms.

Column	Type	Description
contract_id	UUID	Primary key
binding_id	UUID	FK → bindings.binding_id (UNIQUE, NOT NULL)
catalog_id	UUID	FK → price_catalog.catalog_id (pricing reference)
rental_period_type	TEXT	DAILY MONTHLY YEARLY (NOT NULL)
rental_duration	INTEGER	Number of periods (e.g., 3 months) (NOT NULL)
gross_amount	DECIMAL(10,2)	Total rental amount before split (NOT NULL)
platform_fee_percentage	DECIMAL(5,2)	Platform commission rate (NOT NULL)
platform_fee_amount	DECIMAL(10,2)	Calculated platform fee (NOT NULL)
authority_share_percentage	DECIMAL(5,2)	Government authority share rate
authority_share_amount	DECIMAL(10,2)	Calculated authority share
net_owner_amount	DECIMAL(10,2)	Net amount to owner after all fees (NOT NULL)
currency	TEXT	Currency (DEFAULT SAR)
contract_terms	JSON	Additional terms {cancellation_policy, late_payment_penalty}
signed_at	TIMESTAMP	Contract signature timestamp
status	TEXT	DRAFT ACTIVE COMPLETED TERMINATED (NOT NULL, DEFAULT DRAFT)
created_at	TIMESTAMP	Contract creation (DEFAULT NOW())
updated_at	TIMESTAMP	Last update (DEFAULT NOW())

Relationships: 1:1 → bindings | 1:N → ledger_entries | Referenced by: financial_transactions

Table: financial_transactions

Records all payment gateway transactions. Handles TNA issuance fees and rental payments with strict activation blocking until paid status.

Column	Type	Description
transaction_id	UUID	Primary key
user_id	UUID	FK → users.user_id (payer, NOT NULL)
transaction_type	TEXT	TNA_ISSUANCE_FEE RENTAL_PAYMENT SERVICE_FEE (NOT NULL)
reference_type	TEXT	Type of related entity: TNA_REQUEST BINDING RENT_CONTRACT
reference_id	UUID	ID of related entity
quoted_amount	DECIMAL(10,2)	Amount from price_catalog quote (NOT NULL)
paid_amount	DECIMAL(10,2)	Actual amount paid
currency	TEXT	Currency (DEFAULT SAR)
status	TEXT	QUOTED PENDING PAID FAILED REFUNDED PARTIALLY_REFUNDED (NOT NULL)
payment_method	TEXT	CREDIT_CARD DEBIT_CARD MADA APPLE_PAY BANK_TRANSFER
gateway_transaction_id	TEXT	Payment gateway reference (UNIQUE)
gateway_response	JSON	Full gateway response payload
payment_metadata	JSON	Additional payment details {installment_plan, promo_code}
paid_at	TIMESTAMP	Payment confirmation timestamp
refunded_amount	DECIMAL(10,2)	Amount refunded if applicable
created_at	TIMESTAMP	Transaction initiation (DEFAULT NOW())
updated_at	TIMESTAMP	Last status update (DEFAULT NOW())

Relationships: 1:N → ledger_entries | 1:N → settlement_adjustments (for refunds)

Table: ledger_entries

Double-entry accounting ledger for reconciliation and compliance. Every financial action creates balanced debit/credit entries.

Column	Type	Description
entry_id	UUID	Primary key
transaction_id	UUID	FK → financial_transactions.transaction_id
rent_contract_id	UUID	FK → rent_contracts.contract_id (if rental-related)
entry_type	TEXT	DEBIT CREDIT (NOT NULL)
account_type	TEXT	VISITOR_LIABILITY PLATFORM_REVENUE PLATFORM_FEE AUTHORITY_REVENUE OWNER_REVENUE OWNER_ACCOUNT (NOT NULL)
account_reference	TEXT	Account identifier (e.g., OWNER:owner_id, PLATFORM, AUTHORITY)
amount	DECIMAL(10,2)	Entry amount (NOT NULL, POSITIVE)
currency	TEXT	Currency (DEFAULT SAR)
description	TEXT	Human-readable description of entry
reconciliation_status	TEXT	PENDING RECONCILED DISPUTED (DEFAULT PENDING)
reconciled_at	TIMESTAMP	Reconciliation timestamp
fiscal_period	TEXT	Accounting period (e.g., 2026-Q1)
metadata	JSON	Additional ledger metadata {invoice_id, receipt_id}
created_at	TIMESTAMP	Entry creation (DEFAULT NOW())

Relationships: Constraint: Balanced entries for each transaction (sum of debits = sum of credits) |
Updates: owner_accounts

Table: owner_accounts

Accumulates earnings for each property owner. Tracks available balance for payout processing.

Column	Type	Description
account_id	UUID	Primary key
owner_id	UUID	FK → owners.owner_id (UNIQUE, NOT NULL)
current_balance	DECIMAL(10,2)	Available balance ready for payout (NOT NULL, DEFAULT 0.00)
pending_balance	DECIMAL(10,2)	Earnings not yet settled (DEFAULT 0.00)
total_earned	DECIMAL(10,2)	Lifetime total earnings (NOT NULL, DEFAULT 0.00)
total_paid_out	DECIMAL(10,2)	Lifetime total paid to owner (NOT NULL, DEFAULT 0.00)
currency	TEXT	Currency (DEFAULT SAR)
payout_method	TEXT	BANK_TRANSFER WALLET CHECK (DEFAULT BANK_TRANSFER)
payout_details	JSON	Bank account details {bank_name, account_number, iban, swift}
minimum_payout_threshold	DECIMAL(10,2)	Minimum balance required for payout (DEFAULT 100.00) (TIME)
last_payout_at	TIMESTAMP	Last successful payout timestamp
account_status	TEXT	ACTIVE SUSPENDED CLOSED (DEFAULT ACTIVE)
created_at	TIMESTAMP	Account creation (DEFAULT NOW())
updated_at	TIMESTAMP	Last balance update (DEFAULT NOW())

Relationships: 1:N → payout_records | Updated by: ledger_entries (via triggers)

Table: payout_records

Tracks batch payout processes (run-batch). Records fund transfers from owner accounts to registered payout methods.

Column	Type	Description
payout_id	UUID	Primary key
owner_account_id	UUID	FK → owner_accounts.account_id (NOT NULL)
payout_batch_id	TEXT	Batch process identifier for grouping payouts
payout_amount	DECIMAL(10,2)	Amount transferred (NOT NULL)
currency	TEXT	Currency (DEFAULT SAR)
payout_method	TEXT	BANK_TRANSFER WALLET CHECK (NOT NULL)
payout_details	JSON	Transfer details {bank_reference, transfer_date, account_info}
included_ledger_entries	JSON	Array of ledger_entry_ids included in payout [{entry_id, amount}]
status	TEXT	INITIATED PROCESSING COMPLETED FAILED CANCELLED (NOT NULL)
initiated_at	TIMESTAMP	Payout initiation timestamp (DEFAULT NOW())
completed_at	TIMESTAMP	Successful transfer timestamp
failure_reason	TEXT	Reason if status=FAILED
bank_confirmation	TEXT	Bank transfer confirmation reference
processing_fee	DECIMAL(10,2)	Transaction fee charged by payment processor
net_payout_amount	DECIMAL(10,2)	Amount after processing fees
created_at	TIMESTAMP	Record creation (DEFAULT NOW())

Relationships: Updates: owner_accounts.current_balance and total_paid_out

Table: settlement_adjustments

Handles refunds and financial corrections. Fully audited adjustments that affect owner accounts and ledger.

Column	Type	Description
adjustment_id	UUID	Primary key
adjustment_type	TEXT	REFUND CORRECTION PENALTY BONUS CHARGEBACK (NOT NULL)
related_transaction_id	UUID	FK → financial_transactions.transaction_id
related_rent_contract_id	UUID	FK → rent_contracts.contract_id
affected_account_id	UUID	FK → owner_accounts.account_id
adjustment_amount	DECIMAL(10,2)	Amount of adjustment (can be negative) (NOT NULL)
currency	TEXT	Currency (DEFAULT SAR)
reason	TEXT	Detailed reason for adjustment (NOT NULL)
initiated_by_user_id	UUID	FK → users.user_id (who initiated)
approval_required	BOOLEAN	Whether adjustment requires approval (DEFAULT TRUE)
approved_by_user_id	UUID	FK → users.user_id (who approved)
approved_at	TIMESTAMP	Approval timestamp
status	TEXT	PENDING APPROVED REJECTED PROCESSED REVERSED (NOT NULL)
processed_at	TIMESTAMP	Processing timestamp
ledger_entry_ids	JSON	Array of created ledger_entry_ids [{entry_id}]
metadata	JSON	Additional details {original_amount, dispute_id, ticket_id}
created_at	TIMESTAMP	Adjustment creation (DEFAULT NOW())

Relationships: Creates: ledger_entries | Updates: owner_accounts

Financial Money Flow

Step 1: Fee Calculation & Quoting

When a visitor requests a TNA or creates a binding, the system queries price_catalog to generate a quote. The quote considers item_type, pricing_rules, and any dynamic factors (visitor type, volume discounts, seasonal rates).

Step 2: Payment & Order Creation

Visitor pays via integrated payment gateway. A financial_transaction record is created with status=QUOTED → PENDING → PAID. CRITICAL: Binding activation is strictly blocked until transaction status=PAID is confirmed.

Step 3: Settlement & Ledger Entry

Once PAID status is confirmed, the system: (1) Creates rent_contract with gross/net split calculation, (2) Generates balanced ledger_entries for double-entry accounting: DEBIT visitor liability, CREDIT platform revenue, CREDIT authority share, CREDIT owner revenue. (3) Updates owner_accounts.pending_balance.

Step 4: Owner Payout (Batch Process)

Earnings accumulate in owner_accounts.current_balance. Automated batch process (run-batch) executes periodically: (1) Identifies accounts with balance $\geq$ minimum_payout_threshold, (2) Creates payout_records with status=INITIATED, (3) Transfers funds to owner's registered payout method, (4) Updates status=COMPLETED and owner_accounts.total_paid_out.

Step 5: Adjustments & Refunds

Refunds and corrections are handled via settlement_adjustments table. Each adjustment: (1) Requires approval (if adjustment_amount > threshold), (2) Creates offsetting ledger_entries, (3) Updates affected owner_accounts, (4) Links to original financial_transaction for audit trail.

Component	What it does
Rent Contracts	The formal agreement that defines who gets what percentage of the rent.
Ledger Entries	The "Master Record" that proves where every riyal came from and where it went.
Settlement Adjustments	A specialized way to handle refunds or penalties that requires human approval before the math changes.
Owner Accounts	A digital "wallet" for each property owner that tracks their total lifetime earnings and current balance.

DOMAIN 7: Audit & Compliance

Table: admin_audit_log

System-wide immutable audit trail. Every create, update, delete, and export operation is logged with before/after snapshots.

Column	Type	Description
audit_id	UUID	Primary key
actor_type	TEXT	USER SYSTEM API SCHEDULED_JOB (NOT NULL)
actor_id	UUID	FK → users.user_id or system identifier (NOT NULL)
action_type	TEXT	CREATE UPDATE DELETE EXPORT LOGIN LOGOUT PERMISSION_CHANGE (NOT NULL)
target_entity	TEXT	Table name (e.g., visitors, bindings, financial_transactions) (NOT NULL)
entity_id	UUID	ID of the affected record
old_values	JSON	Complete snapshot of record BEFORE action (for UPDATE/DELETE)
new_values	JSON	Complete snapshot of record AFTER action (for CREATE/UPDATE)
ip_address	TEXT	Client IP address
user_agent	TEXT	Browser/client user agent
request_id	TEXT	Request trace ID for correlation
metadata	JSON	Additional context {reason, notes, related_entities}
created_at	TIMESTAMP	Audit entry timestamp (DEFAULT NOW(), IMMUTABLE)

Relationships: None (immutable audit table) | **Constraints:** NEVER DELETE, NEVER UPDATE

Table: user_activity_log

User-facing activity log for dashboards. Tracks meaningful user actions and provides activity history.

Column	Type	Description
log_id	UUID	Primary key
user_id	UUID	FK → users.user_id (NOT NULL)
activity_type	TEXT	TNA_REQUESTED BINDING_CREATED PAYMENT_MADE PAYOUT_RECEIVED etc. (NOT NULL)
Description	TEXT	Human-readable description of activity (NOT NULL)
activity_data	JSON	Related data {entity_type, entity_id, summary}
is_visible	BOOLEAN	Show in user dashboard (DEFAULT TRUE)
created_at	TIMESTAMP	Activity timestamp (DEFAULT NOW())

Relationships: None

DOMAIN 8: System Configuration

Table: issuance_policy

System-wide configuration for TNA issuance rules, fees, and operational policies.

Column	Type	Description
policy_id	UUID	Primary key
issuance_mode	TEXT	MODERATED AUTONOMOUS (NOT NULL)
max_active_tnas_per_visitor	INTEGER	Global limit on concurrent TNAs per visitor (DEFAULT 3)
tna_validity_days	INTEGER	Default TNA validity period in days (DEFAULT 365)
minimum_rental_period_days	INTEGER	Minimum binding duration in days (DEFAULT 30)
eligibility_rules	JSON	Complex eligibility rules {min_age, allowed_nationalities}
is_active	BOOLEAN	Policy active status (DEFAULT TRUE)
effective_from	TIMESTAMP	When policy takes effect
created_at	TIMESTAMP	Policy creation (DEFAULT NOW())
updated_at	TIMESTAMP	Last update (DEFAULT NOW())

Relationships: None (singleton configuration table)

Complete Relationship Map

Core User Relationships

- users (1) → (1) visitors
- users (1) → (1) owners
- users (1) → (1) gov_users
- users (1) → (1) carrier_staff
- users (1) → (N) financial_transactions
- users (1) → (N) user_activity_log

Government & Carrier Entities

- government_agencies (1) → (N) gov_users
- government_agencies (1) → (N) government_agencies (hierarchy)
- carriers (1) → (N) carrier_staff
- carriers (1) → (N) shipments

TNA Lifecycle

- visitors (1) → (N) tna_issuance_requests
- visitors (1) → (N) visitor_tnas
- tna_issuance_requests (1) → (1) visitor_tnas
- gov_users (1) → (N) tna_issuance_requests (as reviewer)

National Address Registry

- owners (1) → (N) owner_national_addresses
- owner_national_addresses (1) → (N) na_sub_addresses
- gov_users (1) → (N) na_sub_addresses (as verifier)

Binding & Rental

- visitor_tnas (1) → (N) bindings
- na_sub_addresses (1) → (N) bindings
- owners (1) → (N) bindings (as approver)
- bindings (1) → (N) binding_status_history

Delivery & Logistics

- visitor_tnas (1) → (N) shipments
- carrier_staff (1) → (N) shipments (as assigned handler)
- shipments (1) → (N) shipment_messages
- shipments (1) → (N) shipment_status_logs

Financial Flows (New Architecture)

- price_catalog (referenced by) → rent_contracts | financial_transactions
- bindings (1) → (1) rent_contracts
- users (1) → (N) financial_transactions

- financial_transactions (1) → (N) ledger_entries
- rent_contracts (1) → (N) ledger_entries
- owners (1) → (1) owner_accounts
- ledger_entries (updates) → owner_accounts
- owner_accounts (1) → (N) payout_records
- settlement_adjustments (affects) → owner_accounts
- settlement_adjustments (creates) → ledger_entries
- financial_transactions (1) → (N) settlement_adjustment

Critical Business Rules Constraints

Financial Rules

- Binding activation BLOCKED until financial_transaction.status = PAID
- Every rent_contract must have balanced ledger_entries (debit sum = credit sum)
- Owner payouts only process when current_balance ≥ minimum_payout_threshold
- All settlement_adjustments require approval before processing

Sub-Address Verification

- No sub-address can be used in binding unless is_verified = TRUE
- Government official verification (verified_by_gov_user_id) is mandatory

National Address Ownership

- Every owner_national_address requires title_deed_reference
- ownership_proof_status must be VERIFIED before creating sub-addresses

Audit Trail

- admin_audit_log is IMMUTABLE - no updates or deletes allowed
- All CRUD operations on sensitive tables must create audit entries